

Ahmad Maraqa *AI Research Engineer*

ahmadmaraka2002@gmail.com | +962 7 9038 5041 | Amman, Jordan | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFILE

AI Research Engineer with 2+ years of experience across AI/ML, large language models, computer vision and AI automation. I work with LLM systems (RAG, embeddings, tool-calling agents), machine learning and deep learning, and self-taught n8n automation, having independently built and shipped 30+ production workflows. I apply these technologies to solve real-world problems and deliver impactful, high-quality solutions that people can rely on.

EDUCATION

10/2020 – 09/2024 Amman, Jordan	Computer Engineering <i>University of Jordan</i> GPA: 3.77 (Excellent)
07/2014 – 07/2020 Amman, Jordan	High School Diploma <i>Al-Raed Al-Arabi School</i> Scientific Stream (98.1%)

PROFESSIONAL EXPERIENCE

01/2025 – present Amman, Jordan	AI Research & Integration Engineer, Penguin Location Services ● Develop LLM applications using RAG, embeddings, vector databases, and local/open-source LLMs via Ollama. ● Apply prompt engineering to build AI-powered assistants that integrate language models with external tools. ● Research and evaluate AI/ML approaches through trial and error, keeping what holds up on real data rather than on a clean test set. ● Work with BLE, IoT devices, controllers and MQTT in RTLS applications, applying machine learning models to noisy sensor data.
10/2024 – 01/2025 Amman, Jordan	AI Research Engineer Intern, Penguin Location Services Worked with BLE and IoT devices in RTLS applications, training AI models (KNN, Decision Trees, Random Forest) and implementing a custom probabilistic model during the internship.
07/2024 – 09/2024 Amman, Jordan	AI and ML Engineer Intern, Leading Point Software Onsite internship in data science, machine learning, deep learning and computer vision: developed machine learning models including image classification and object detection, with hands-on experience of AutoML tools.

INDEPENDENT WORK

Ongoing Independent	AI Automation & Agents, Self-Taught / Independent ● Learned n8n independently and built 30+ production workflows, including tool-calling Telegram agents (task management, voice-in/voice-out scheduling, CV review, customer support with human handover). ● Built a production WhatsApp Business Cloud API integration backed by MongoDB, with verified webhooks and rate-limited batch sending. ● Self-host and maintain n8n on a Hostinger VPS; built a self-updating RAG knowledge base over Pinecone.
------------------------	--

PROJECTS

Dublejli – AI-Driven Video Dubbing System

Exhibited at SOFEX 2024 (Aqaba) and National Technology Parade 2024. Integrated Speech-to-Text, Translation, and Text-to-Speech, with a custom synchronization algorithm to match the dubbed audio to the original video.

Google Calendar Voice Assistant (n8n)

Voice/text Telegram assistant that creates, reads, updates, and deletes Google Calendar events via four dedicated tools, confirming details before saving; replies in text or speech to match the input.

WhatsApp Business Cloud API Integration (n8n, MongoDB)

Production client messaging system with MongoDB as source of truth, verified webhooks, and rate-limited batch sending.

Self-Updating RAG Knowledge Base (n8n, Pinecone)

Google Drive–synced RAG pipeline (Pinecone + HuggingFace embeddings) that ingests only new files on a schedule.

Jordanian Currency Detection and Counting Using YOLOv8 [GitHub](#) | [Live Demo](#)

Custom-collected dataset for detecting and counting Jordanian notes and coins; deployed as a live Streamlit app.

Fire & Smoke Detection – Binary Classification on Sensor Data [GitHub](#) | [Live Demo](#)

Trained seven ML models (Logistic Regression through a Keras neural network) on 62K+ IoT sensor readings; deployed a Streamlit app comparing all seven side-by-side.

TECHNICAL SKILLS

AI/LLMs: RAG, Embeddings, Vector DBs (Pinecone, FAISS), Prompt Engineering, Agentic Workflows, Tool Calling, Ollama, LangChain

Machine Learning & CV: TensorFlow, Keras, Scikit-learn, OpenCV, YOLO (Ultralytics), Pandas, NumPy

Automation & Integration: n8n (self-hosted), Python, Telegram/WhatsApp/Gmail/Google APIs, MongoDB, Webhooks, REST APIs

Engineering: Java, C++, Verilog, Git/GitHub

LANGUAGES

• Arabic (Native) • English (Professional)